

UNMASKING SAMPLING BIAS: *PhyloThin* REVEALS THE TRUE DIVERSITY OF PROKARYOTIC GENOME COLLECTIONS



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More and More Data ...

in NCBI database, 402 bacteria & archaea species have more than 10 complete genome entries

BUT ...

How many genome-samples are meaningful?

NCBI: 27,070 samples → *PhyloThin* → actually only... 16,081 effective samples

(NCBI data until August 4, 2023)



- estimate the "effective sample number" (identify sampling bias and non-effective samples)
- based on coalescent theory

Input: inferred phylogeny

Goal: identify genomes that represent a "more randomly sampled" subset

Benefits:

- reduce the amount of data to analyze with minimal loss of diversity
- avoid biased predictions

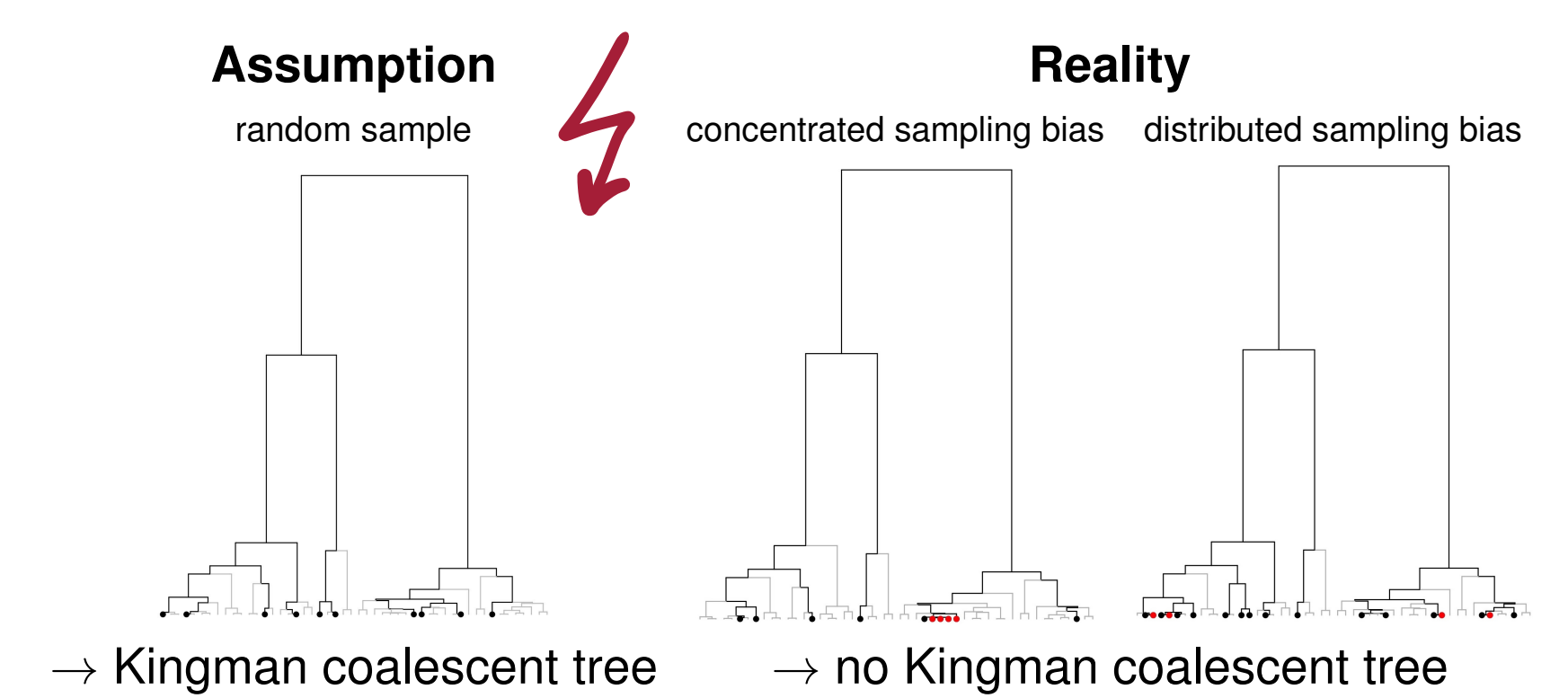
Strong Sampling Bias ...

- reduces the effective information in databases
- underestimates bacterial diversity, adaptive potential and the prevalence and spread of traits such as antibiotic resistance

Warning for Machine Learning Methods based on Public Prokaryotic Genome Collections

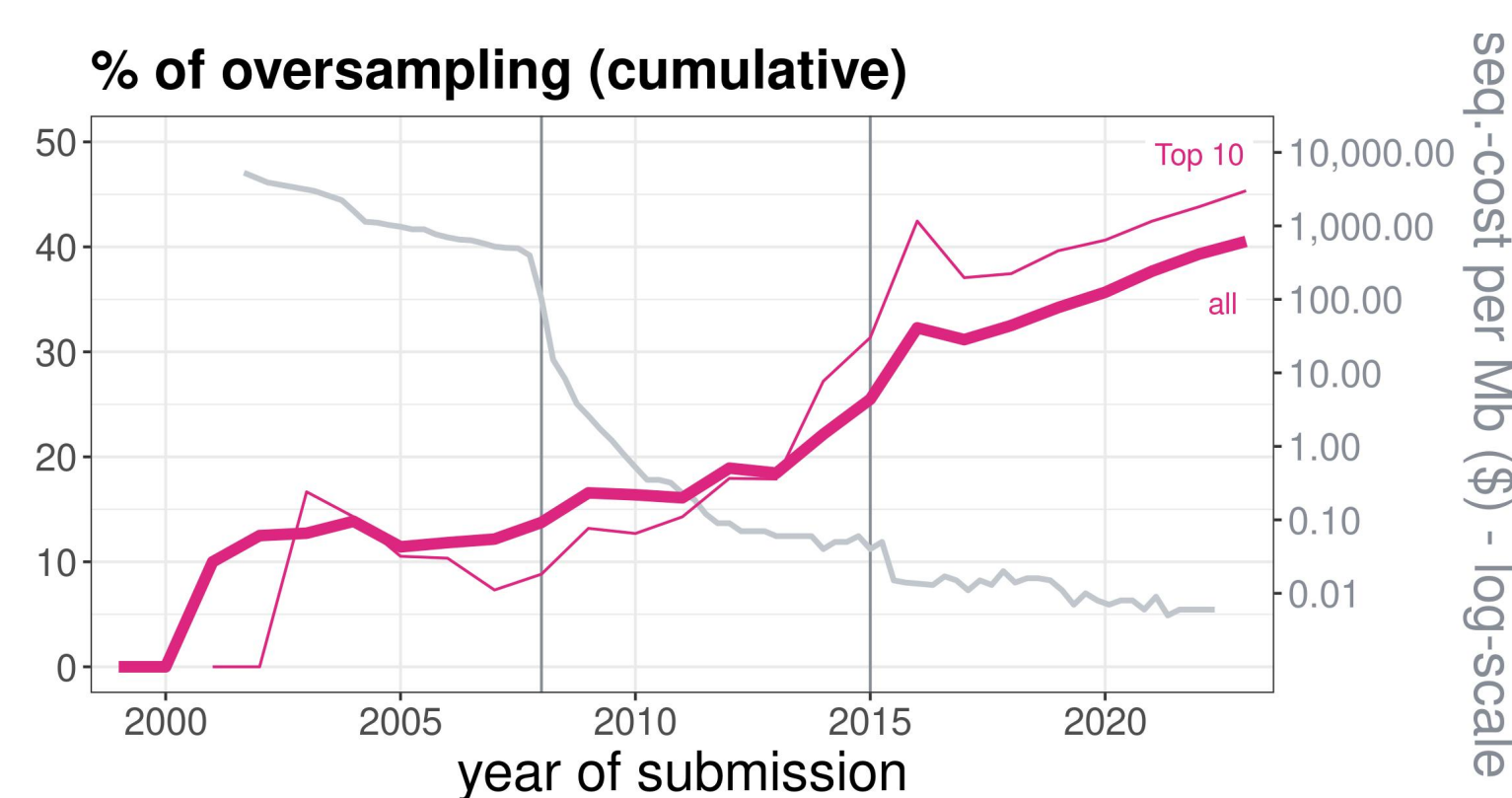
If not removed, the existing strong sampling bias...

- will reduce the effective information in databases
- can result in misleading conclusions including machine learning applications

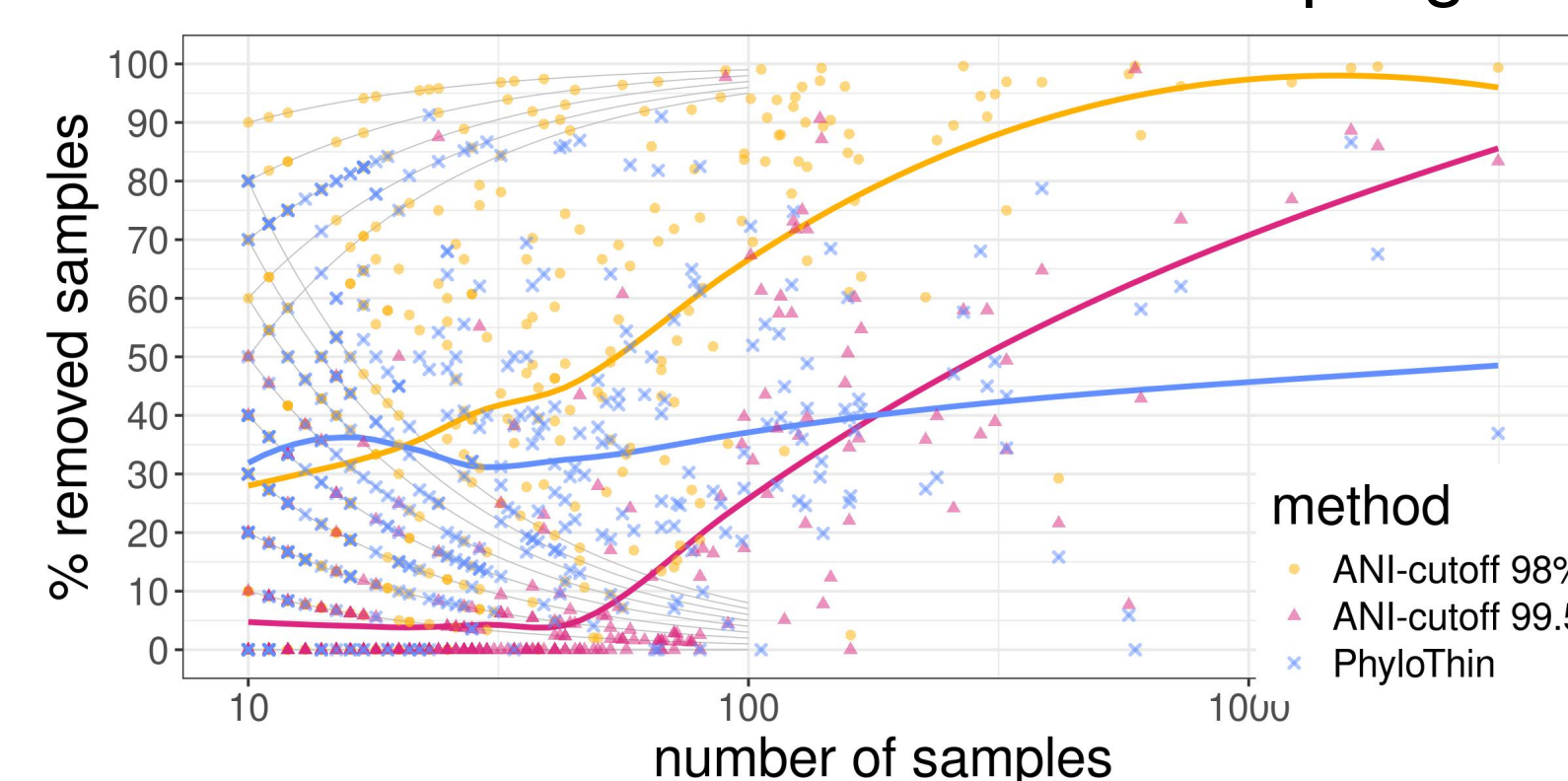


- *supervised machine learning*: simulated training data usually unbiased, sampling bias in real data will distort inference
- *unsupervised machine learning*: sampling bias in training data propagates to hidden internal features

Sampling Bias is a Widespread and Increasing Problem



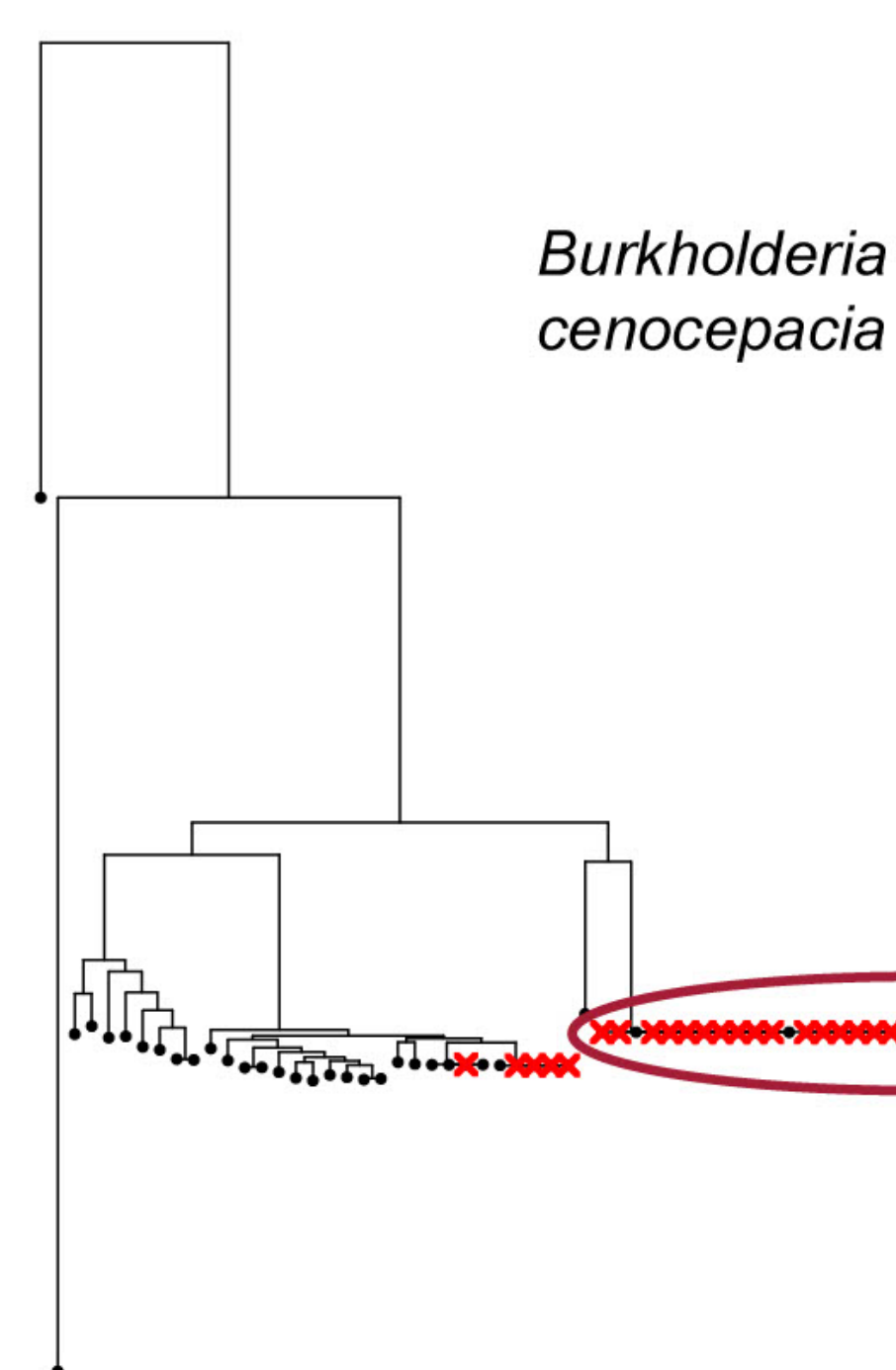
ANI-threshold does NOT remove sampling bias



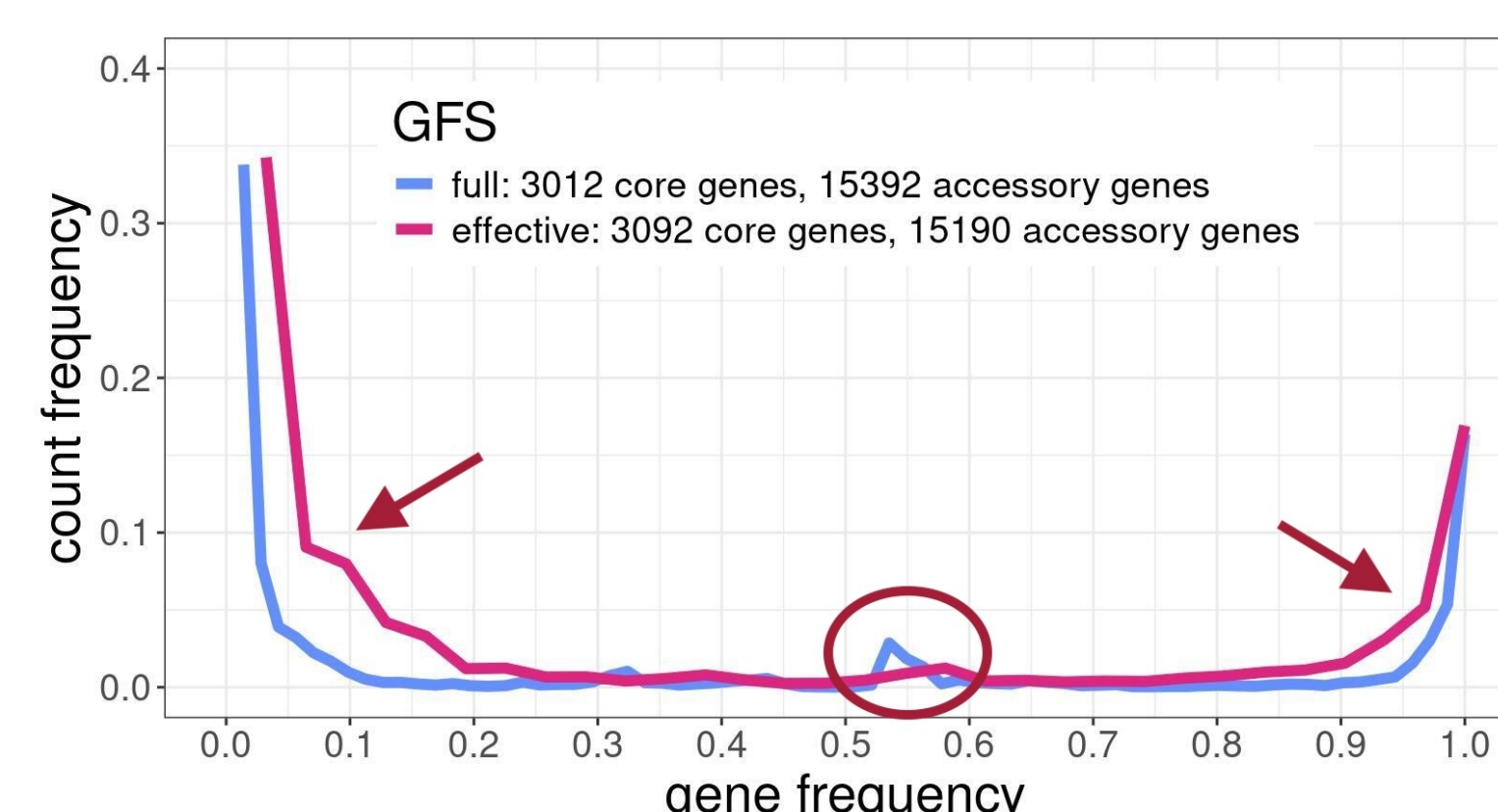
	full dataset	n	oversampling		effective dataset	n_e	p_{100}	g_1
1	<i>E. coli</i>	3152	36.9% ±0		<i>Escherichia coli</i>	1988	1.6%	6.0
2	<i>K. pneumoniae</i>	1811	67.5% +2		<i>Staphylococcus aureus</i>	1174	2.2%	2.0
3	<i>S. enterica</i>	1602	86.6% +4		<i>Bordetella pertussis</i>	593	0.4%	0.2
4	<i>S. aureus</i>	1219	3.7% -2		<i>Klebsiella pneumoniae</i>	588	5.6%	17.8
5	<i>P. aeruginosa</i>	732	62.0% +3		<i>Mycobacterium tuberculosis</i>	542	3.8%	2.2
6	<i>A. baumannii</i>	609	58.1% +3		<i>Helicobacter pylori</i>	351	6.5%	2.1
7	<i>B. pertussis</i>	593	0.0% -2		<i>Pseudomonas aeruginosa</i>	278	12.9%	36.1
8	<i>M. tuberculosis</i>	576	5.9% -2		<i>Acinetobacter baumannii</i>	255	14.0%	29.1
9	<i>H. pylori</i>	417	15.8% +2		<i>Enterococcus faecium</i>	215	11.8%	13.1
10	<i>L. monocytogenes</i>	386	78.8% -7		<i>Salmonella enterica</i>	214	16.7%	43.6

Adjustment for sampling bias reveals that the best-covered species are not the ones with the most sequenced genomes. The table shows results for the full species datasets (left) and effective species datasets after thinning (right) the phylogenetic trees based on core genes with *PhyloThin*. For each species, the observed sample size is given alongside the species name. The percentage of oversampling is given for full species datasets. The metric p_{100} denotes the predicted increase in genes after unbiased sampling of another 100 genomes and the metric g_1 the predicted number of new genes expected from sequencing the next unbiased sampled genome.

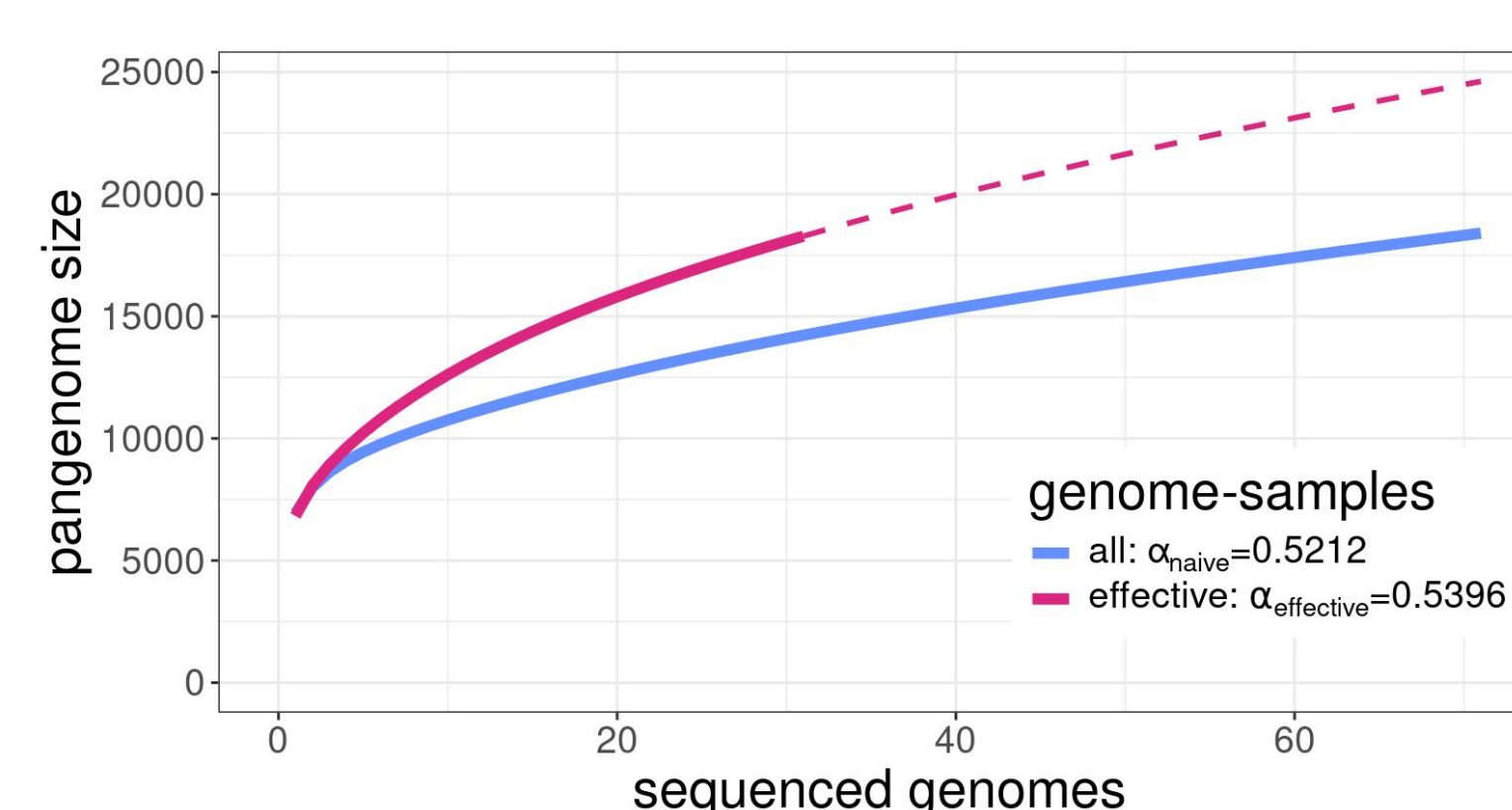
Core-Genes Phylogeny



Gene Frequency Spectrum



Pangenome Accumulation Curve



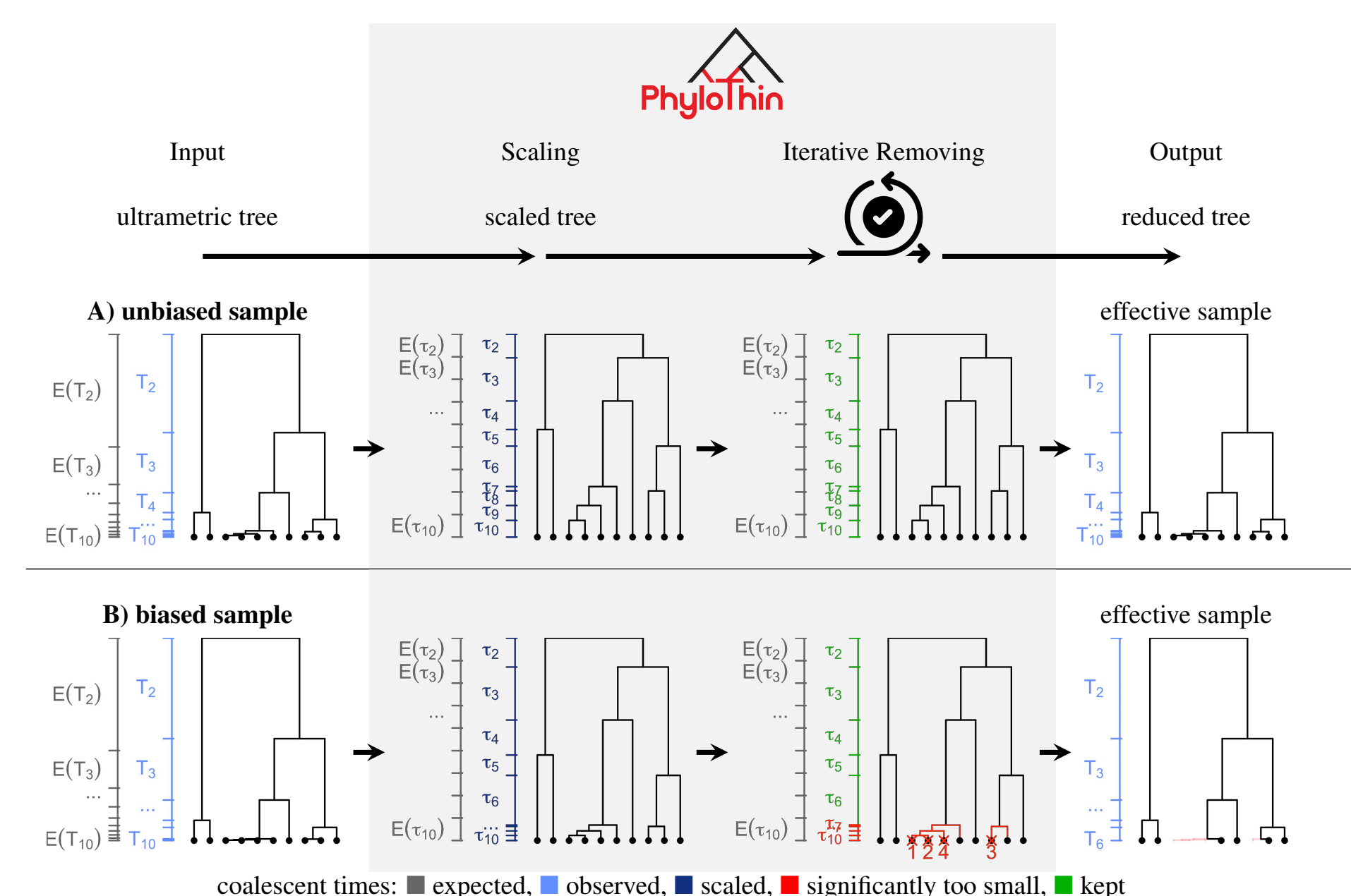
Sampling bias can obscure the real distribution of SNPs and gene frequencies and in the case of oversampling often leads to an excess of genes in intermediate frequency. Moreover, sampling bias can flatten pangenome accumulation curves, leading to an underestimation of the diversity of the species.

(NCBI data until August 4, 2023)

Conclusion

Public prokaryotic databases contain substantial sampling bias that continues to increase. With *PhyloThin*, we present a fully automated statistical tool to **detect and correct for sampling bias in prokaryotic populations**. Removing the strong sampling bias allows to identify the effective amount of information and prevents misleading biased conclusions including the diversity and adaptive potential of bacteria. In addition, unmasking sampling bias helps to prioritize species where additional sampling would most effectively capture microbial biodiversity in future sequencing efforts.

How does *PhyloThin* work?

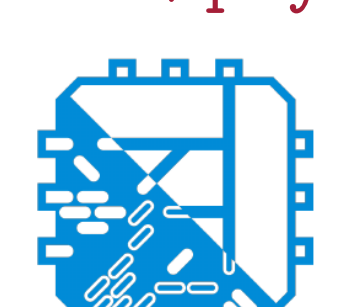


- under neutrality and without sampling bias, **coalescent times** can be **scaled** to be Exp[1]-distributed
- with sampling bias, external coalescent times are significantly **shorter** than expected
- *PhyloThin* tests scaled external branches and **iteratively removes** significantly too short external branches
- *PhyloThin* also offers two additional methods: *Mutation-rate-sensitive Thinning* for low mutation rates and *Subsampling* for large sample sizes

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github.com/fbaumdicker/phylothin



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